# SHAFT DESIGN COMPLIANCE REPORT

**Standard:** ASME Code recommendations for Transmission Shafting
**Material Selected:** EN19 (Alloy Steel)

## 1. Design Input Parameters

| Parameter | Value | Description |
|---|---|---|
| Power (P) | 20.0 kW | Transmitted shaft power |
| Rotational Speed (N) | 960.0 RPM | Operational speed |
| Bending Moment (M) | 520,000.0 N-mm | Peak bending load |
| Fatigue Factors (Kb / Kt) | 1.7 / 1.2 | ASME combined shock/fatigue modifiers |
| Specified FOS | 1.0 | Design margin multiplier |
| Keyway Cutout Included | Yes (25% Stress Reduction Applied) | Keyway geometry check |

## 2. Mechanical Properties (Database Reference)

| Property | Value |
|---|---|
| Yield Strength (S_yt) | 520.0 MPa |
| Ultimate Tensile Strength (S_ut) | 700.0 MPa |
| Modulus of Elasticity (E) | 200.0 GPa |
| Density | 7850.0 $kg/m^3$ |
| Design Allowable Shear Stress (t_allow) | 94.50 MPa |

## 3. Design Calculations & Geometric Output

| Output Metric | Value | Equations / Notes |
|---|---|---|
| Calculated Torque (T) | 198,958.3 N-mm | T = (9550 * P / N) * 1000 |
| Equivalent Twisting Moment (Te) | 915,673.3 N-mm | $Te = sqrt((Kb*M)^2 + (Kt*T)^2)$ |
| Shaft Profile Type | Solid | Structural geometry option |
| Analytical Minimum Diameter | 36.68 mm | Calculated stress boundary limit |
| <b>Recommended Standard Diameter</b> | <b>40 mm</b> | Rounded to nearest ISO standard profile |

## 4. Secondary Dynamic & Deflection Analysis

| Analysis | Value | Comments |
|---|---|---|
| Max Induced Shear Stress (tau) | 72.87 MPa | Induced stress under service load |
| Computed Margin / FOS | 2.14 | Margin against material shear limit |
| Est. Total Deflection | 0.4342 mm | Combined load & weight contribution |
| First Critical Speed | 1,435.4 RPM | Estimated fundamental resonant frequency |
| Bearing Reaction forces | 2080.0 N | Simplified symmetrical support model |

## Design Verification Statement:

The recommended outer diameter of **40 mm** satisfies the analytical requirements under the applied torsional and bending conditions with a margin of safety. Real-world fatigue, environmental effects, and key stress concentration zones must be independently validated.